

Mean Absolute Deviation

Lesson 8-3

Name: _____

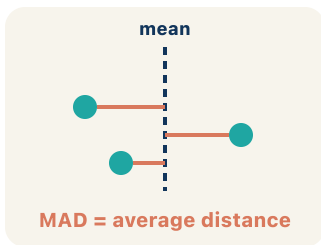
Date: _____

Class: _____

Key Vocabulary

Level 1 support

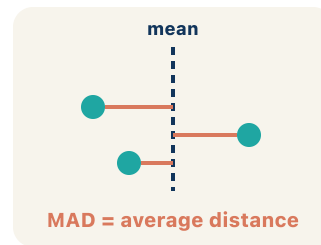
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Data: 8, 10, 12. Mean = 10. Distances from mean:
2, 0, 2. $MAD = (2+0+2) \div 3 = 1.33$

Mean Absolute Deviation

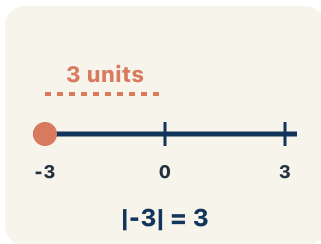
The average distance of each number from the mean.



If mean = 20 and value = 17, deviation = $17 - 20 = -3$ (3 below the mean)

Deviation

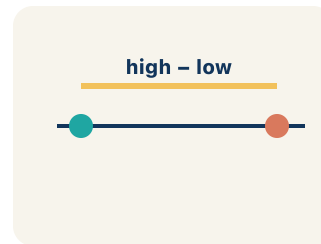
How far a number is from the mean.



$|-3| = 3$ and $|3| = 3$ — both are 3 units from zero

Absolute Value

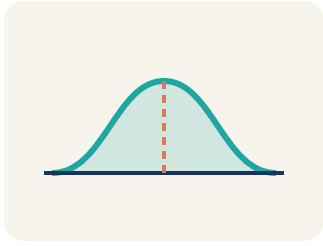
How far a number is from zero. It is always positive.



Low spread: 8, 9, 10, 11 (close together). High spread: 2, 9, 10, 25 (far apart)

Spread

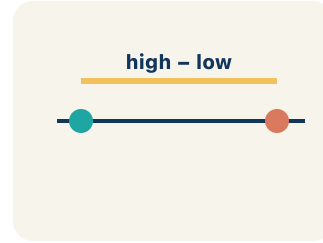
How far apart the numbers are.



A set clustered tightly around the mean has low MAD; a set spread far from the mean has high MAD

Data distribution

How the data looks: where it sits and how spread out it is.



Low variability (MAD = 1): very consistent. High variability (MAD = 8): very spread out

Variability

How spread out the numbers are.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find the mean absolute deviation (MAD) to describe how spread out data is.

1 Notice

Coach wants to know which basketball player is more consistent: Player A scored 18, 22, 20, 24, 16 points in the last 5 games.

2 Key idea

I can find the mean absolute deviation (MAD) to describe how spread out data is.

3 Words I'll use

Mean Absolute Deviation

Deviation

Absolute Value

Spread

Data distribution

Variability



Think About It

- Both players have the same mean. What is different about their scores?
- Which player's scores stay closer to 20?
- Which player has the biggest single-game difference from 20?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

I do — watch

Follow each step as your teacher solves it.

Problem: The mean of a data set is 15. One value is 11. What is the absolute deviation of that value from the mean?

- A. 4
- B. -4
- C. 26
- D. 11

Step 1 Deviation = $11 - 15 = -4$.

Step 2 Absolute deviation = $|-4| = 4$.

Answer: A. 4

 We do — together

Solve this one with your class using the same steps.

Problem: A data set has absolute deviations of 3, 1, 5, 2, 4. What is the MAD?

- A. 3
- B. 5
- C. 15
- D. 1

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: The mean of a data set is 20. A value is 26. What is the absolute deviation?

- A. 6
- B. -6
- C. 46
- D. 26

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Player A scored 18, 22, 20, 24, 16 and Player B scored 10, 30, 25, 12, 23. Both average 20. Which player would you count on to score close to 20 every night, and why?

Level 1 support

Start here: Two data sets can share a mean but differ in how spread out they are.

 **Need a hint?**

Sentence starters (optional)

I would count on Player ____ because their scores are ____.

Contaría con el Jugador ____ porque sus puntajes están ____.

Even with the same mean, the players differ in ____.

Aunque tengan la misma media, los jugadores difieren en ____.

Word bank: mean absolute deviation deviation spread consistent mean

Listen for: Students choose Player A as more consistent and recognize that equal means hide different spreads.

Level 2

Predict: which player will have the larger MAD, and what does a larger MAD say about consistency?

- Player ____ will have the larger MAD because ____.
- A larger MAD means the scores are ____, so the player is ____.

EXPLORE

To find the MAD for Player A (18, 22, 20, 24, 16, mean 20), why do we take the ABSOLUTE VALUE of each deviation before averaging?

Level 1 support

Start here: Absolute value makes every distance from the mean positive so they don't cancel out.



Need a hint?

Sentence starters (optional)

We use absolute value so the deviations are all ____.

Usamos el valor absoluto para que las desviaciones sean todas ____.

Without absolute value, the deviations would add up to ____.

Sin valor absoluto, las desviaciones sumarían ____.

Word bank: mean absolute deviation deviation absolute value spread mean

Listen for: A strong answer explains deviations above and below the mean would cancel to zero, so absolute value keeps each as a positive distance.

Level 2

Compute and justify: find the MAD of 18, 22, 20, 24, 16. Show the deviations, take absolute values, and average them.

- The absolute deviations are ____, which sum to ____.
- Dividing by 5 gives a MAD of ____, meaning ____.

CONNECT

Two keepers both allow a mean of 1.67 goals, but Keeper A's MAD is 0.56 and Keeper B's is 1.22. Which keeper should the coach start in the championship, and why?

Level 1 support

Start here: A smaller MAD means more consistent, predictable performance.



Need a hint?

Sentence starters (optional)

The coach should pick Keeper ____ because their MAD is ____.

El entrenador debe elegir al Portero ____ porque su DAM es ____.

A smaller MAD means the keeper is more ____.

Una DAM más pequeña significa que el portero es más ____.

Word bank: mean absolute deviation deviation spread consistent mean

Listen for: Students pick Keeper A and explain a smaller MAD (0.56) means goals allowed stay close to the mean, so performance is more reliable.

Level 2

Critique: when might a coach actually prefer the LESS consistent keeper (higher MAD)? Defend with a scenario.

- A coach might prefer the higher-MAD keeper when ____.
- Higher spread could be acceptable if ____ because ____.

REFLECT

For the data set 4, 8, 6, 10, 2 with a mean of 6, talk through the steps to find the MAD.

Level 1 support

Start here: Find each deviation from the mean, take absolute values, then average them.

 **Need a hint?**

Sentence starters (optional)

First I find each deviation from 6, then take the ____.

Primero hallo cada desviación de 6, luego tomo el ____.

I average the absolute deviations to get a MAD of ____.

Promedio las desviaciones absolutas para obtener una DAM de ____.

Word bank: mean absolute deviation deviation absolute value spread average

Listen for: Students compute absolute deviations 2, 2, 0, 4, 4 (sum 12), divide by 5, and report MAD = 2.4.

Level 2

Generalize: what does it mean about a data set if its MAD is exactly 0?

- A MAD of 0 means every value is ____.
- That happens because each deviation from the mean is ____.

Write About the Math

The Writing Revolution

I can explain my work using the words mean absolute deviation, deviation, absolute value, and spread.

1. Kernel Sentence subject + verb

Model: Mean Absolute Deviation is the average distance of each number from the mean.

Desviación media absoluta es la distancia promedio de cada número a la media.

Write a kernel sentence about mean absolute deviation. Use a subject and a verb.

Escribe una oración base sobre desviación media absoluta. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Mean Absolute Deviation matters in math

Desviación media absoluta importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Mean Absolute Deviation matters in math because ____.

Desviación media absoluta importa en matemáticas porque ____.

but
pero

Mean Absolute Deviation matters in math, but ____.

Desviación media absoluta importa en matemáticas, pero ____.

so
entonces

Mean Absolute Deviation matters in math, so ____.

Desviación media absoluta importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about mean absolute deviation.

Di un hecho verdadero sobre mean absolute deviation.

Mean absolute deviation ____.

Question *Pregunta*

Ask a question about mean absolute deviation.

Haz una pregunta sobre mean absolute deviation.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about mean absolute deviation.

Muestra entusiasmo sobre mean absolute deviation.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with mean absolute deviation.

Dile a un compañero qué hacer con mean absolute deviation.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I would count on Player ____ **because their scores are** ____.

Contaría con el Jugador ____ *porque sus puntajes están* ____.

Even with the same mean, the players differ in ____.

Aunque tengan la misma media, los jugadores difieren en ____.

We use absolute value so the deviations are all ____.

Usamos el valor absoluto para que las desviaciones sean todas ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. The data 10, 10, 10, 10 has what MAD?

- A. 0
- B. 10
- C. 2.5
- D. 40

Show your work:

2. For the data 2, 4, 6 (mean 4), what is the MAD?

- A. 1.33
- B. 2
- C. 4
- D. 12

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two basketball teams both average 75 points per game. Team A's last 5 scores: 73, 76, 74, 77, 75. Team B's last 5 scores: 60, 90, 65, 85, 75. Calculate the MAD for each team and explain which team a coach would prefer if they need predictable scoring.

Sentence starter: Team A's MAD = _____. Team B's MAD = _____. Team A is _____ because _____. A coach would prefer Team _____ for predictable scoring because _____.

Show your work:

Reflect — Exit Ticket

Data set: 4, 8, 6, 10, 2. The mean is 6. What is the MAD?

- A. 2.4
- B. 6
- C. 0
- D. 12

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 3 — *Sum of absolute deviations* = $3 + 1 + 5 + 2 + 4 = 15$. $MAD = 15 \div 5 = 3$.
2. **You Try (notes):** A. 6 — *Deviation* = $26 - 20 = 6$. *Absolute deviation* = $|6| = 6$.
3. **Try It 1:** A. 0 — *Every value equals the mean, so each distance is 0 and MAD = 0*.
4. **Try It 2:** A. 1.33 — *Distances are 2, 0, 2; mean* = $4 \div 3 \approx 1.33$.
5. **Exit Ticket:** A. 2.4 — *Deviations: -2, 2, 0, 4, -4. Absolute deviations: 2, 2, 0, 4, 4. Sum = 12.*
 $MAD = 12 \div 5 = 2.4$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Mean Absolute Deviation is the average distance of each number from the mean.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).